6.2 Gauss's 17-gon

WORKSHEET

The question

STEP II 1991, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why does the first part, about $\cos\frac{2\pi}{5}$, appear in a question about the 17-gon?

STAGE 1 The pentagon

Covers: the first part, $\phi = \frac{2\pi}{5}$

Ask yourself Expand $y^2 = (\cos\phi + \cos 2\phi)^2$ and turn each product into a sum with $\cos A \cos B = \frac{1}{2}\big(\cos(A+B) + \cos(A-B)\big)$.

Pause and try Use $5\phi = 2\pi$ to reduce $\cos 3\phi$ and $\cos 4\phi$. Which root of the quadratic is the right one?

STAGE 2 The seventeenth roots of unity sum to zero

Covers: the sum of the cosines for $\theta = \frac{2\pi}{17}$

Ask yourself $\sum_{k=0}^{16} \cos k\theta$ is the real part of a geometric series in $\omega = e^{i\theta}$. What is its sum?

Pause and try Use $\cos(17-k)\theta = \cos k\theta$ to pair up the terms.

STAGE 3 Splitting the cosines into two groups

Covers: the value of z

Ask yourself Let z' be the sum of the other four cosines, $\cos 3\theta + \cos 5\theta + \cos 6\theta + \cos 7\theta$. What is $z + z'$?

Pause and try Work out zz': expand it as 16 products, turn each into two cosines, and reduce the angles. How often does each of $\cos\theta, \ldots, \cos 8\theta$

appear?

STAGE 4 The quadratic

Covers: the value of z

Ask yourself z and z' are the roots of a quadratic. Which root is z? Estimate the four cosines in z.

What's really going on

Gauss periods

Why $\{1, 2, 4, 8\}$? Modulo 17, the powers $3^0, 3^1, 3^2, \ldots$ of 3 run through all the non-zero residues:

$$1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6.$$

Taking every other one, $3^0, 3^2, 3^4, \ldots$, gives 1, 9, 13, 15, 16, 8, 4, 2, which are ± 1, ± 2, ± 4, ± 8 modulo 17: exactly the angles in z. The other half gives $\pm 3, \pm 5, \pm 6, \pm 7$, the angles in z'. Gauss called the sums of ω^k over such classes *periods*. Because $17 - 1 = 16 = 2^4$, the residues can be halved four times, and at each stage the periods satisfy quadratic equations whose coefficients involve the periods of the previous stage. Four quadratics,

so four square roots, lead from the rational numbers down to $\cos\frac{2\pi}{17}$:

$$\cos\tfrac{2\pi}{17} = \tfrac{1}{16}\Big(-1+\sqrt{17}+\sqrt{34-2\sqrt{17}} \\ +2\sqrt{17+3\sqrt{17}-\sqrt{34-2\sqrt{17}}-2\sqrt{34+2\sqrt{17}}}\Big).$$

Every step involves only square roots, and square roots can be constructed with ruler and compasses (using similar triangles or a semicircle), so the 17-gon is constructible.

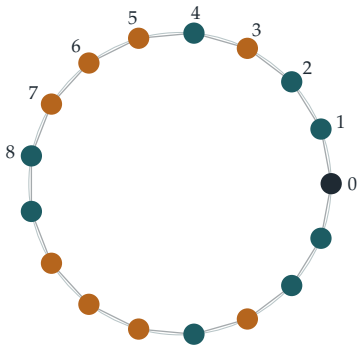


Figure 6.2.1 The vertices of the regular 17-gon, split into Gauss's two periods. The teal vertices are at angles $\pm1, \pm2, \pm4, \pm8$ (in units of $\frac{2\pi}{17}$), and the sum of their x-coordinates is $2z = \frac{-1+\sqrt{17}}{2}$; the amber vertices give $2z' = \frac{-1-\sqrt{17}}{2}$.

Which polygons can be constructed?

Gauss claimed, and Pierre Wantzel proved in 1837, that a regular n-gon can be constructed with ruler and compasses exactly when n is a power of 2 times a product of distinct *Fermat primes*, primes of the form $2^{2^m}+1$. The known Fermat primes are 3, 5, 17, 257 and 65537, the Fermat numbers of Section 1.1. (Fermat guessed that all such numbers are prime, but Euler showed that $2^{32}+1$ is divisible by 641, and no further Fermat primes have ever been found.) So the 257-gon and the 65537-gon are constructible; in the nineteenth century Johann Hermes spent about ten years working out the construction of the 65537-gon, and his manuscript is kept at Göttingen. The 7-gon and 11-gon are not constructible, because 7 and 11 are not Fermat primes ($7-1=6$ and $11-1=10$ are not powers of 2), and neither is the 9-gon, because $9=3^2$ repeats the Fermat prime 3. Wantzel's proof uses the theory of field degrees from Section 6.3.

Try this next

1 **The heptagon.** Try STEP II 1995, Question 6 (step.maths.org/questions/95-s2-q6). It applies Gauss's idea of grouping roots of unity to the seventh roots, and finds exact values for sums of cosines and sines. Deciding the sign of the sine sum is the subtle

step.

2 **The pentagon.** Show that $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$, and use it to describe a ruler-and-compasses construction of a regular pentagon.

3 **The other period.** Check numerically that $z' = \cos 3\theta + \cos 5\theta + \cos 6\theta + \cos 7\theta = \frac{-1-\sqrt{17}}{4}$.

4 **Generators.** Show that the powers of 3 modulo 17 give every non-zero residue, but the powers of 2 do not. Which residues do the powers of 2 give?

Further reading

- **Ian Stewart, *Galois Theory*** (CRC Press). Chapter on ruler-and-compass constructions and the regular 17-gon.
- **Richard Courant & Herbert Robbins, *What Is Mathematics?*** (Oxford University Press). Chapter 3 is a beautiful elementary account of constructions, including the 17-gon.
- **Daniel Kehlmann, *Measuring the World*** (Quercus). A comic novel about Gauss and Alexander von Humboldt, for light relief.

The full guided solution is in Section 6.2:
https://beyondstep.keithmcnulty.org/ch06/gauss-17gon.html